CPU Scheduling ready → run → no special timer mechanism

1. non-preemptive: process 主动 give it up when terminated / waiting I/O / 线程 yield statement

preemptive: ① Time Mechanism ② waiting/non-waiting give it up ③ priority — user / kernel mode

④ complete I/O or some special event

dispatch latency: 开始真 current process 中断 → kernel → next process (进入OS, 准备信息 context switching)

preemptive: ③ ④ ⑤

non-preemptive: ① ② ④ → 会 convoy effect: ① ③ ④

starvation: ② ③ ④

2. Scheduling Algorithm

① FCFS ② STF ③ SRT ④ PR ⑤ RR ⑥ Multi-level (Feedback) Queue

CPU utilization, Throughput, Turnaround Time (完成-到达), Waiting Time, Response Time

(六) Multi-Level Queue Scheduling 个 ready queue ⇒ 多 seperate queues

System: Priority Interactive: RR ⇒ response time

Batch: FCFS, SRT, STF ⇒ Turnaround Time

SRT, STF, 抢/非抢, 都会最优

average wait/turnaround

队列划分标准 (2种): ① Fixed Priority Scheduling ⇒ starvation ② Time slice

① 作业性质划分 ② 作业不会一直执行 high priority

(七) Multi-Level Feedback Queue process 可以在 queue 间 upgrade/downgrade

queue 内部 scheduling algorithm, queue 间也有 (上述①②)

Memory Management

1. CPU 可直接访问内存 (register, cache, MM)

2. Address Binding

(1) Compile time: process 的 address 在 确定有这个 process 时就被固定 VP = MP

(2) Load time: process 的 address 在 disk → MM 后, 被固定 VP = MP → VP = MP 不同 bandwidth

(3) Run time: process 的 address 在 执行时, 都可以变化 VP ≠ MP 不是 disk

↳ OS + VM 实现的基础

→ 都经历过 Load, 所以 base + offset, process 大小不可超过 Limitation

3. Memory Allocation

low memory: Resident OS; high memory: User process

(1) Contiguous —— Fragmentation 碎片: 用 compaction 解决 MUT 的内部碎片

(2) Non-Contiguous

① Paging: PTBR: 放 base register, TLB: 快查表 ⊗ 同时放 虚拟的 page/frame 编号

② Segment: external fragment

Virtual Memory

Demand Paging ⇒ FIFO, LRU, Optimal Anticipatory Paging ⊗ 预取分页

2种 trap { reference is invalid: terminal; page is not in MM: wait }

effective virtual access time = (1−f) × EAT + f × page fault service time

EAT (effective access time) = t(1 + memory access time) + (1−t)(2 + memory access time) t = hit, f = not hit

Additional reference bit algorithm 每隔固定时间, shift right, 换 value 最大的

Second chance algorithm ① 1 → 0 (二次机会) ② 找第一个 0 的, 查 1 → page ③ 直到找到 1 个 0 (t = 0) 替换

Fixed allocation { Equal; Proportional } ⇒ Local page replacement

Variable allocation ⇒ Local & Global page replacement

Thrashing: Process are busy with doing swap page ⇒ CPU utilization ↓

due to 少量 frames ⇒ 大量 page fault, I/O blocking ↑

内存利用率 进程数量

① local replacement

② global replacement 会放大这种影响 Page fault frequency (PFF)

Working-set: 保证足量的 frame

Δ − 段 time 内, process exchange 总量

size: Δ 内, process 种类数 > frame size ⇒ 增加 frame

单 process PF ⇒ 增加 frame

若大部 process 需要 PF ⇒ 选择一部 suspend

FIFO: Belady's anomaly

Deadlock request → ready queue → scheduling → run → release

Deadlock Characterization

① Mutual exclusion: 互斥, 同一时间, 只有 1 个 process 可用 resource

② Hold and Wait: 在等待 other resource 时, 手里不许有 resource ⇒ resource utilization ↓, starvation

③ no preemptive

④ Circular wait: priority order

4 methods to deal with deadlock

① Ostrich approach ② D prevention (预防)

③ D avoidance (提前分配 resource) ⇒ use additional infor

i. 事先声明对每种 resource 的最大需求数

ii. 动态检查资源分配后, system 是否仍处于 safe state

iii. 资源分配之状态: 由可用资源数, 已分配资源数和 process 最大需求数共同决定

④ D detection: Process Termination: ① 几切 ② 1个1切 ⇒ starvation

(检查 + 恢复) Process Rollback: 重生后可能又走死在上一次死亡点

如果 request edge 得到许可, 就会 reverse direction ⇒ assignment edge

消除有向边判断 deadlock ⇒ 从 Pi 入手, 若其 assignment / requesting edge 可以得到满足,

我们使 Pi 等 P 的 both requesting / assignment edge

算法本质有向图 若存在 cycle, 拒绝相符 一定无 cycle 才相符

Cooperating System (被) 影响有集合内其他 process

Synchronization 保证 cooperation 的有序进行 randomly choose / cicular queue

Critical Section Problem

① Mutex Exclusion: 互斥: 会发生 collision → 只有它有性, 不影响他程正常跑

② Progress (CS 不能空着, 不该有 process 可进) → 证明: someone wants to enter, 其中有人能进

③ Bounded Waiting: 证明: 工 请求后, 一定可以进 (有限 time 内 = without forever)

int count = 0, int in = 0, int out = 0, semaphore S = 1	
Producer	***Consumer***

while (true) do {
// produce an item and store it
while (count == NUM_BUF) do { };
// wait for an unused buffer
P(S);
buffer[in] = nextProduced;
in = (in + 1) % NUM_BUF;
count = count + 1;
V(S);
}
	while (true) do {
	while (count == 0) do { };
	// wait for an available item
	P(S);
	nextConsumed = buffer[out];
	out = (out + 1) % NUM_BUF;
	count = count − 1;
	V(S);
	// consume a stored item
	}

P(S)
Disable Interrupt
S.value = S.value − 1
if (S.value < 0)
block (S.queue)
Enable interrupt

V(S)
Disable Interrupt
S.value = S.value + 1
if (S.value <= 0)
wakeup(S.queue)
Enable interrupt

while (true) do{
while (TestAndSet(lock)) do{};
<CS>
lock = false
<RS>}

boolean TestAndSet (V){
boolean returnvalue = V;
V = true;
return returnvalue}

monitor 长的像 semaphore 结构

外部的 single queue ⇒ P(mutex)

中间只允许 1个 process 执行 ⇒ CS

signal { Java: continuous → 主动 wait; standard: wait → 唤醒 }

wait queue ⇒ conditional value

wait ⇒ V(); P()

signal ⇒ V(mutex)

File Concept → disk 上的 smallest unit: file 是 abstract data type (ADT)

(1) file is a named collection of related information 有名字的相关信息集合

(2) file infor − 一般在 directory structure 上,

(3) file system: a collection of files with proper storage and directory structure

(4) 1个 disk 可有多个 partition (file system), 但1个 partition 也可由多个 disk 组成

(5) attribute: Name, Identifier, Type, Location, Size, creator/owner, Protection, Time

(6) file operation: Create, Read, Write, Reposition, Delete, Truncate (清空)

Open ① 从目录结构中找到该文件 ② load from disk to MM; Close ① store from MM to disk

① file pointer ② file-open count ③ Disk location ④ access right

只有当最后1个 process 用完关闭 file, 才得出计数 ↳ 记录 file 绝址, avail 有没有结构目录开辟太大

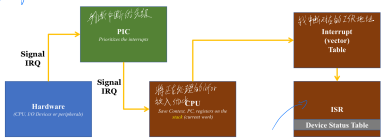
(7) 2 major types of files ① Program files: Source code, Object code, Execution Program

② Data file: Character or ASCII file, Binary file, Free-formatted file, Formatted or structured

③ { unstructured: 但单 word/bytes → Unix; structured: Simple record structure (固定 − 拼 − 拼), complex structure (html) }

Unix 的内核只接收 bytes 字符流, 不支持结构化 file, 但是由程序自己负责 (结构实现)

Magic Number → help with it



IRQ – Interrupt Request
PIC – Programmable Interrupt Controller (prioritizes interrupts, and ensures the CPU responds to the most critical ones first)
ISR – Interrupt Service Routine or Interrupt handler
PC – Program Counter

Access Method

Sequential Access: access in order, 支持 read/write next, reset, skip forward/backforward

Direct Access (relatively access), access directly, 支持 Read/Write n 用 Position n + Read/Write next 定位

Index Access: 建立 separate index file (包含 pointers to data block), data is stored in direct/relative file

Directory Structure —— 本身就 file, a collection of nodes containing infor of all files

manager file directory (Search, Create, Delete, List, Rename, Traverse 遍历)

1. Single-Level Directory: Name Problem (2 文件不可同名), Group Problem (无法文件分类)

2. Two-Level: MFD → UFD, seperate directory for each user, 不同 user 间可有 same name 文件

faster searching, access rights —— user 去命令到 以的 file

3. Tree-Structured: multiple levels, more efficient, grouping capacity 分类分组

current working directory: absolute or relative path names

4. Acyclic-Graph: 允许1个 file/sub-dire 被多个 sub-dire 共享 (aliasing 别名)

Problem: ① double count ② Dangling Pointer —— 删 link, file 也没了, 其他 link 变无效

Solution: reference count 创/删 link 时 +1/−1, 直到计数 = 0, 才删 file

通过任意别名 (aliasing) 修改 file, file 同步更新

5. General-Graph: directory-link-directory 形成 cycle, reference count 失效

① garbage collection ② i. 定期检查 ii. 每日要不要并有 link/path ii. 每次清除新link 相符 重新拉起

File Sharing —— ① multi-user system 多用户共同处理同一文件, protection scheme (认证)

② distributed system (use network)

③ remote file system (Manually (FTP, winscp), Automatically (J drive), 半自动 (web))

1. Client Server: 客户端挂载 mount 远程文件系统, 调用命令去文件, 调用接口, 远程执行并返回结果

2. Network-file system (NFS): Unix/Linux 系统下, 最常用的分布式文件系统 (iCloud, dropbox, Google drive.)

3. Storage-Area Network (SAN): 大量数, 大容量, 高速

4. Cloud-Storage: Over the cloud:

File Storage Allocation

1. Linked Allocation: sequencial access 2. Indexed Allocation: random access

Storage 1. Secondary Storage 2. Disk Storage (local, remote) 3. Terbely 三级

① Host-Attached Storage (most standard arrangement) local, improve fault tolerance —— mirror hard disk (RPID 0+1)

② Network-Attached (NAS): remote procedure call Unix NFS − NAS based

③ Storage-Area Network (SAN): 当 file-sharing, private data network, special storage protocols

3. Primary: MM, 直连 CPU Secondary: hard disk/SSD, I/O bus Tertiary: CD-ROM, DVD

↳ hierarchical storage structure ① 用于 backup & archival 归档

② Robotic optical jukebox, collection of cartridge ③ offline storage: Isolate 隔离等

Protection & Security → OS 的 kernel code → rule

1. Protection: mechanisms & policy exi 修改资源, 限制访问 Internal OS

2. Security: 涉及用户认证, 消息加密到 malicious (恶意) 时意外的 introduction or flow (漏洞) to external OS

Problem: Intruders/crackers (入侵者) 企图 breach (突破) security

threat (威胁): 潜在安全威胁 (violation)

attack: 企图 breach (突破) security (可以是 accidental or malicious)

更好防护

Security Violation Categories

1. Breach of confidentially: 机密性破坏: 未授权读取 data (严重危害)

2. Breach of integrity: 完整性破坏: 未授权改动 data (严重危害)

3. Breach of availability: 可用性破坏: 未授权销毁 data (严重危害)

4. Theft of service: 服务盗用: 未授权使用 data (严重危害)

⑤ Denial of service (DOS): 拒绝服务攻击, 阻止合法用户正常使用服务, 给系统负担时攻击 server

} security goal

Breaching Security 4种安全攻击方式

1. Masquerading (伪装成别) 假充合法用户, 获取权限和权限

2. Relay attack (重放攻击): 把正确数据包记录下 重新发送 server, 欺骗目标

3. Man-in-the-middle attack (中间人攻击): 在两个 data flow 中, 冒充 sender or receiver (命令信息 "公共密钥中转机")

4. Session hijacking (会话劫持) ex: 劫持已有登录状态的, 跳过身份验证 (登录) 权限

Program Threats 5种 (程序被安全威胁: (code 后面的 后患, malicious 恶意)

1. Trojan horse 木马: 隐藏恶意代码伪装成合法程序, 的恶意意图 "公共密钥中转机" 伪装机制 ex: 伪装成 login 登录, 窃取密码

2. Trap door 后门: 开发者/设计者 预留的特殊入口, 可以绕过正常安全检查 快速

3. Logic bomb 逻辑炸弹 平时正常运行, 一旦满足某特定 condition (ex 某个日期), 就执行恶意程序

4. Stack & buffer overflow 栈与缓冲区溢出, 利用 program code 漏洞, 写入超出缓冲区限制的 data, 覆盖 stack 上的 return 地址, 调用 return, 跳转到恶意代码

5. Viruses 病毒 嵌入合法程序 code 中的片段, 具有自我复制能力, 破坏目标数据 系统 computer

通常针对特定 CPU 架构, OS, app, 常见传播方式: email, macro

Security Measure Level 4个 安全防护 level

1. physical level ex: 机房门禁 锁与钥 devices

2. Human level ex: 员工培训 防骗

3. OS level ex: file right, process isolation, bug fixed

4. Network level ex: 防止 通信 被窃听, 防火墙

Computer Security Classification 安全分级

D: Minimal security C: discretionary protection 基于单元的访问保护

主体用保护模式的操作用户访问 → 访问控制

主体用权限列表的操作用户访问 → 存储访问控制

B: 包含 C 的全部特性, 每个 object 都有安全等级 label

A: 采用形式化设计与验证技术 来确保安全 (最高级)

Information Security CIA

confidentiality / Integrity / Availability

non-repudiation

不可否认性: 保证了行为者 (会被记录)

Cryptography ⟷ 与 cryptography 互补

1. Transposition cipher 换位置排, 按规律变换

2. Substitution cipher 用密钥表, 字母替换

3. Vigenere cipher 循环增加

4. Caesar's cipher 每个字母都移动固定位数 (A → D, D → B)

Authentication

Knowledge / Ownership / Inherence factor

2 (multi-) factor authentication

certificate: public-key infrastructure

① possession of private key 持有私钥 (公钥加密信息)

② 持有权威机构签发并数字签名的证书

Window Scheduling

Window 的 dispatcher 采用 preemptive priority-based RR 调度器 —— 32 不同的 priorities

real time (16-31), user class (1-15), Memory Management (0)

6个 priority class 高级 REALTIME — IDLE-Priority Class

7个 relative classes TIME-CRITICAL — IDLE

用户或 I/O 的 process 获得的 响应时间 会提 提升 前台进程 (正在用的窗口) 时间片更长 (比后台进程)

普通 process 默认 属于 Normal-Priority-Class

Linux Scheduling

140个 priority classes (100 real-time jobs, 40 others)

priority value 动态调整 取决于交互情况 real-time (static), other jobs (dynamic)

却被 匹配 → next process → all processes 用完 匹配 → all 重置

Memory Management

1. Pentium Memory (IA-32) (at most 16384 segments) { 2-level paging 4KB; 1-level paging 4MB }

2. ARM Memory VP = MP = 32 bit { 2-level paging 4KB; 1-level paging 1MB }

3. Itanium Memory VP = 48 bit MP = 32 bit 4: 4KB 3: 2MB 2: 1GB

Virtual Memory

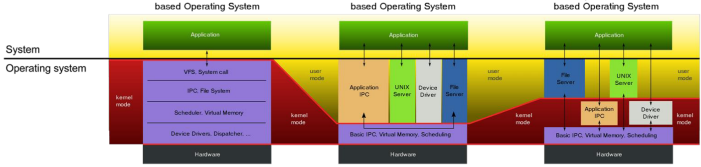
Window Virtual Memory —— demanding paging with clustering → anticipate surrounding page

发生 page fault: System 取 page 可能会 会涉及周围 page

进程在 MM 中, 不可保持 working set minimum 1 page, 不可 分 1X 个 page

若系统 低于 阈值 会回收 超出 minimum 的部分

单 CPU 采用 balanced clock algorithm, 多 CPU 采用 FIFO



Linux Virtual Memory —— demand paging → valid/dirty/accessed bit
1个process 1个page table, 不同process可share MM
page finding 机制: kernel swap daemon thread; LRU aging mechanism

Window & *nix Deadlock 检查的overhead远不及deadlock出现频率
Ostrich approach: 直接ignore deadlock

File System → variation of linked allocation
1. DOS File System (FAT): All links are stored in seperate table 在disk开头
2. Window ↙ 继承
(1) HPFS: 可存入data & special attribute; FNODE, not first cluster as in FAT
(2) NTFS: journaling/recoverable file system by keep track of transaction ⇒ 小变复
Unix hard link; Linux access by open source NTFS server
3. Unix indexed allocation —— Fast file system
file infor 都在 inode中, 不是 file structure
4-level index block —— direct/indirect/double indirect/triple indirect block → large file
4. Linux —— Virtual File System 4 main object
① Superblock: global information ② Dentry: 每文件名和Inode的指针
③ Inode: meta file (每目录结构中内容), 并指向 block ④ File: actual opened file object, an Inode together with a current/writing offset; point to dentry

Synchronization
1. *nix
Message Passing: 普通进程间; Shared Memory: 同1进程中的多线程 Pthread库
Critical Section —— ① 用 mutex-locks ② Conditional variables
③ barrier 所有人到齐后, 再一起往下执行
2. Window
Critical Section —— ① 用 mutex/semaphore (special critical section) ④ barrier
② Conditional variables ③ 排列 wait function (synchronization objects)
Access Control —— access control list
1. *nix
(1) 单 process → 多 domain (2) /etc/passwd ✗ ⇒ authentication
(3) all objects in Unix are files (硬件也被抽象成 file)
(4) access control rwx —— owner/group/other
2. Window
(1) An item can be a file or a directory and has an owner. → can change control list, 此别人用
(2) 相较于 *nix 的 rwx, 还用了 D: delete, P: modify ACL → 直接剥夺所有权
owener/user 得到允许可改ACL D: make current account the new owner
(3) Packaging of access privileges into permission sets ④ full control
to simplify control: ① no access ② read: RX ③ change: RWXD ④ RWXDPO

Recently Development
(1) Microsoft: open source (2) Win 10: Windows Subsystem full Ubuntu kernel under
(3) Win 11: Powershell → Windows Terminal

1. Input/Output Direction M keyboard/screen → 改到想要的 (ex: file)
- There is a second output place for programs, called the standard error. It is normally used only for special system error messages and the standard error is also the screen.

> 清空原文件再写入 / 无原文件, 创建新文件再写入
>> 追加 $ GPA < grade2.txt (&) >> transcript.txt
- Input read from grade2.txt and output and error appended to transcript.txt

$ prog1 < infile | prog2 > outfile
- Effect like prog1 < infile > tempfile1; prog2 < tempfile1 > outfile (sequentially)

2. Pipe (concurrently) 在 Unix/Linux 中, 我们将 pipe 视为 I/O redirection, 但实际不同
Concurrency is about independent computations that can be executed in an arbitrary order with the same outcome. The opposite of concurrent is sequential, meaning that **sequential** computations depend on being executed step-by-step to produce correct results.

OS operation interrupt 在 OS 中相当重要
1. ① Dalmost all OS are interrupt-driven → OS 受中断驱动 (keyboard)
② interrupt allows OS to get control of system ex. 除数出现错误
* signal processing for interrupt { no response: terminate; yes: special response }
{ Pc <ctrl><Alt><Del>; Window <ctrl><c> } In Unix: clock algorithm — timer handler
2. interrupt security limitation 中断处理的安全问题
① 若 disabling interrupt (禁用中断), OS can't take back CPU
OS 不能干预 error (死循环), system 会 hang-up (ex: 蓝屏) → 防止用户滥用 interrupt-related 操作
3. ① hardware 的 privileged command, executed by OS & its core component
限制①, 保护 system 安全 security/stability → only be executed
4. dual-mode, ② 的具体实现 user 获取 OS 服务 'trap' 通一合法方式
user mode ⇒ normal (no privileged instruction) instruction | kernel mode ⇒ all instruction

System call user can't use privileged instruction
only OS processes could be executing in kernel mode

1. System Program
① OS ② Device Driver ③ Utility Program
④ Compiler ⑤ Assembler
2. Application Program
① Word Processor ② Spreadsheet
③ Web Server ④ Media Player ⑤ Email Client

3. 5 types
① File System: open(), close(), read()
② Process Control: fork(), exec()
③ Memory Management/Information maintenance brk(), mmap()
④ (Inter process) Communication: pipe(), socket()
⑤ Device Management: setConsoleMode()
user execute system call ⇒ trap
CPU switch ⇒ deal with the trap

① File manipulation ② Status infor ③ File Modification ④ Communication
⑤ Programming Language support ⑥ Program loading & executing ⑦ Utilities

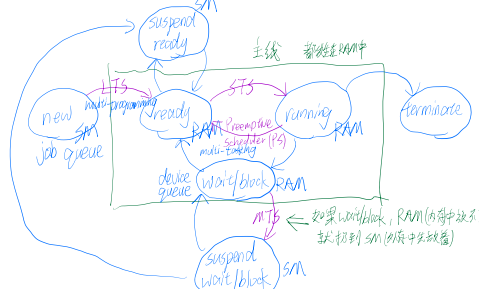
OS type & structure
1. ① Batch programming: 1任务 1任务 执行 Unix, Linux, Window Server
② Time-sharing: Interleaved manner by CPU, but each has time slice
③ Distributed: multiple-computers, network OS ⇒ { current: 单CPU; parallel: 多CPU; distributed: 多comp }
④ Multi-programming: Interleaved manner by CPU Unix
⑤ Real-time: deadline & penalty
2. *nix { Unix { BSD; System V by AT&T; Linux: LAMP development slot-arm } } Linux ⇒ Fastest supercomputer; Apache; MySQL; PHP Perl Python
3. Advantages of Unix & Linux: ① Reliability ② Security ③ Speed ④ Cost ⑤ Open Source

OS structure
1. Unix/Linux (Monolithic Kernel) ① 是一坨, 速度快 ② 耦合性强 ③ 体积大
2. Mac OS X (Microkernel) ① slow, IPC等信息沟通 ② modular 方便添加 ③ 只保证主要部分 Window
Advantages: ① easier to extend a microkernel
② easier to port OS to new architecture/hardware
③ More reliable (less code running in critical kernel mode)
④ More secure
Disadvantages: Performance overhead of user space to kernel space communication
3. Hybird Kernel —— Monolithic 与 Micro 的混合版 (Mac OS)
4. Modular OS —— 本质是 Monolithic, 但有 modular (Linux)
① Object-oriented approach | 1 fail may lead all fail
② Each core component is separated
③ Each talks to the others over known interfaces
④ Each is loadable as needed within the kernel

user — Application Programs — Shell — System Program — System Call — kernel — OS — HW

```
#!/bin/bash ("$@:1:4") ("$@:5")
teams=("$@") ← 将input存储成array
declare -A points ← 新array points
for t in "${team[@]}", do          字符串当下标
    points["$t"]=0  → 初始化 array points   字典
done ⇒ 每个sum=0
for file in round*.txt, do
    if [! -f "$file"]; then continue; fi;
    while IFS= read -r line || [-n "$line"]; do 每1行遍历
        read -r t1 t2 s1 s2 <<< "$line"  划分 → 匹配team name
        if [[ "${team[$t1]}" = ~ "$t1" || "${team[$t2]}" = ~ "$t2" ]]; then
            if ["$s1" -gt "$s2"]; then
                points["$t1"] = $((points["$t1"]+3))
                points["$t2"] = $((points["$t2"]+3))
            elif [ ]; then
            else
            fi
        fi
    done < "$file"
done
for t in "${team[@]}"; do
    echo "$t with ${points[$t]} points"
done output
```

1. 对于 ls -l 以可编辑形式复开 (no hidden)
接受参数 { -a 全部文件名 (包括隐藏文件); -al (全部 + 权限) }
① 类型 { - 文件; d 目录; l 软连接; b 块设备; c 字符设备 }
comp 1412 目录; lab1 可执行文件; link_to_file 符号链接; notes.txt 普通文件
file { symbolic link = 快捷方式; hard link = 副本 }
删除 file { symbolic link: 不会连; hard link: 还存在 }
② -o in cc ⇒ to specify the output filename
将源代码文件, 生成可执行的文件名 cc lab1.c -o lab1 ⇒ / lab1
③ alias (别名) 生成 ⇒ alias ls='ls -FC'
2. chmod bb4 filename ⇒ ① 文件 file; ② director
chmod g-r f
group 失去 read command
user, group, other r=4 w=2 x=1
⇒ 显示结果中, 若 filename 后有 *, 那则一定是 X 权限
3. grep 文件搜索 command
grep -i/-h/-l "text in file" filename
-i: ignore 大小写
-h: 不显示文件名, 只显示相关内容
-l: 只显示文件名
① ② ③
② 'polyu': 以 polyu 开头
'polyu$': 以 polyu 结尾
'poly.': . 可表示任何字符
[abc]: only one of a, b, c
[^abc]: 除了 a, b, c
'^polyu.$'
4. $wc filename ⇒ 4 可结果, 展示该 file 中 line, word, byte, character 数量
5. File Command
$ file filename
$ file filename1 filename2 filename3 ... ⇒ 查看 filenameX 的信息
$ lab*files=lab*.c
$ file $labfiles
$ usefiles="lab*.c *.doc *.pdf"
$ file $usefiles
⇒ 列出符合列出名字的 ___ 相关的文件



① SM → RAM: LTS, 从new中要符合筛选, 放入RAM中
RAM → RAM: STS, 几乎无任何筛选, 追求一个"快"字 (CPU switch)
RAM → SM: MTS, 几乎无任何筛选, 追求一个"快"字 (CPU switch)
↳ thrashing (抖动): 大量时间用于内存换页, 而非执行任务 尽可能多的将 process 放入 RAM
② Type of OS
(1) Multiprogramming: non-preemptive: 除非 process 自己不想干了, 尽可能多的将 process 放入 RAM
(2) Multi-tasking: preemptive: 其他更高级 process 抢了他的 CPU
③ Type of Process LTS makes good use of it
{ I/O-bound process: 长时间等待 I/O, 要插入短时间 CPU 使用
CPU-bound process: 长时间占用 CPU 短时间
↳ 进程的这2种 type 并非固定不变, 而是可切换, 动态变化

2. Cooperation, ① information sharing ② Computation speed-up ③ Modularity ④ Convenience
3. Terminated
(1) zombie process (僵尸子进程): 结束, 但父进程未调用 wait() 进行回收 子进程
(2) orphan process (孤儿): 父进程已结束, 但子进程还在继续
↳ 使用 Daemon Process (守护进程)
4. exec: path, NULL → execv; execlp: file, NULL → execvp; *argv[] → execv: *envp[]

一、Unix/Linux Interprocess Communication
↳ Shared Memory Mechanism
1. IPC Package set up Data Structure in Kernel Space
① 安全性与防御
② kernel 统一管理 all process
③ 不相关 process 也可通过 kernel 进行通信
1. Communication Method
(1) Direct Communication (like 邮箱)
(2) Indirect Communication (like mailbox)
2. Synchronization
Blocking 可以保证不会发生
① lost message ② waste resource: 非最精力看 message 是否到来
2. Share Memory is not practical in Unix/Linux
① 将 data 放入 address 可能会覆盖原 data, 导致 system crash
Protection Mechanism: OS 当中介, data 不直接访问 memory
② thread within the same process naturally share the memory space
User process use system call (Pthread) create shared memory segment

(2) Message Passing Mechanism
one direction
{ pipe: unidirectional, FIFO, unstructured (data stream) kernel 内置 buffer, 点对点, no bandwidth
socket: like named pipe (file) } ① 字节 byte 形式 ② byte flow
二、Communication Topology
① star topology (n-1), bottleneck 2 rounds ② Ring topology (n+1) many rounds
③ Linear topology (n+1), many rounds ④ Fully connected $\frac{n(n-1)}{2}$ fast communication
⑤ Tree topology (n-1), more rounds ⑥ Mesh topology $\frac{nn-1}{2}$ need routing, flexible
⑦ Hypercube: flexible → 2(n-1) ~ n(n-1) ⑧ Bus, not considered with pipes
三、(un)named pipe (本质上是1个file, 但一般用socket替代, not common in Unix/Linux Prog)
用 mkfifo 创建 named pipe, 实际是1个file, 不同process和找到这个file ⇒ heterogenous programs
named pipe is more flexible, while unnamed pipe is more secure
① only parent-child ② no name, can't find in system ③ dead after process dead
unnamed pipe 数量受限: 因为 fd 最大 1024, fd=0,1,2
Wildcard ⇒ normally used in ls
*: 代替任何1个/1组 字符
?: 代替任何1个字符
[]: 限制一个范围 [1-2] 中的1个字符 [0-2] [0-2A-Z] → [1-2] 无法匹配 12 而 {12} 可以
{}: ex: ls lab{1,12,12}.c 精确匹配这三个文件